A CHARACTERISTIC-TWO $\mathbb{Q}$-FACTORIAL THREE-DIMENSIONAL PLT PAIR WITH A NON-NORMAL DIVISORIAL CENTRE

ARGUS OPEN RESEARCH PROJECT

Abstract. We document the counterexample of Cascini and Tanaka showing that in dimension three and characteristic $p = 2$, a $\mathbb{Q}$-factorial purely log terminal (plt) pair (Z, E) with E a prime divisor can have E non-normal.

1. Main Refutation

Theorem 1.1 (Cascini–Tanaka)**.** *Over an algebraically closed field of characteristic $p = 2$, there exists a projective $\mathbb{Q}$-factorial three-dimensional variety Z and a prime divisor $E \subset Z$ such that the pair (Z, E) is purely log terminal (plt), yet E is not normal.*

Date: August 27, 2026.
2020 *Mathematics Subject Classification.* Primary 14E30; Secondary 14G17.